PLASMA PHYSICS & TOKAMAK CONFINEMENT — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Plasma Physics & Tokamak Confinement

Question 1

10.0 points

Examine the toroidal and poloidal magnetic field geometry in the tokamak cross-section shown in the diagram below. Describe how these fields collaborate to confine plasma, and identify any potential areas of instability within the configuration.

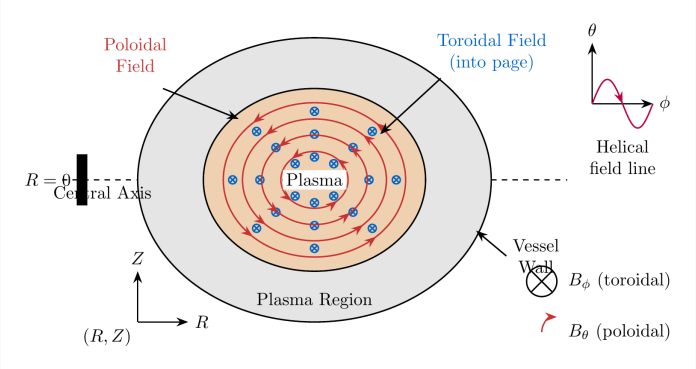

Fig. 1. A diagram of a tokamak cross-section showing the toroidal and poloidal magnetic field geometry. The tokamak is a doughnut-shaped device, and the cross-section should illustrate the circular path of the toroidal field lines around the central axis of the tokamak, as well as the poloidal field lines that loop around the plasma vertically. The diagram should also indicate the plasma region confined by these magnetic fields, and potential areas of instability, such as regions where the fields might be weaker or more prone to disruptions. Label the toroidal and poloidal fields clearly.

Question 2

10.0 points

Analyze the Lawson criterion diagram below that displays the triple product $nT\tau$ versus temperature. Explain how this diagram is used to evaluate the feasibility of fusion reactions.

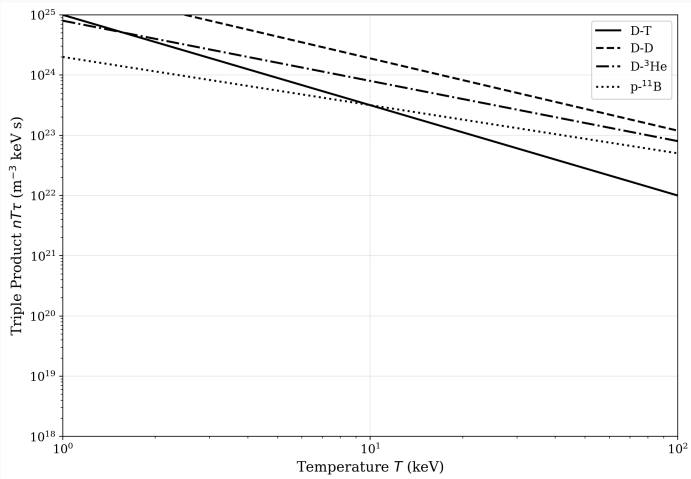

Fig. 2. A Lawson criterion plot showing the triple product nTτ versus temperature. The plot includes labeled axes for nTτ and temperature, with regions indicating the feasibility of different fusion reactions. The diagram helps visualize the conditions required for sustaining fusion reactions, such as the minimum nTτ needed for various temperatures. IMPORTANT correction needed: The diagram description states it should show ". Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 3

10.0 points

Examine the diagram below illustrating magnetic field curvature and grad-B drift for a charged particle in a tokamak. Identify how these drifts affect particle confinement and propose mitigation strategies.

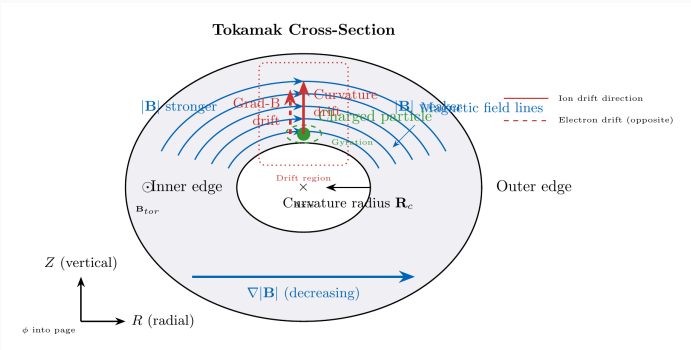

Fig. 3. A diagram illustrating magnetic field curvature and grad-B drift for a charged particle in a tokamak. Show the tokamak as a toroidal structure with magnetic field lines curving around it. Include a charged particle moving along these lines, indicating the direction of the grad-B drift. Label the magnetic field lines, the particle, and the drift direction. Highlight the curvature of the magnetic field lines and the resulting drift path of the particle. IMPORTANT: The diagram is missing required labels. The diagram is missing a required label for. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 4 10.0 points

Examine the q-profile (safety factor) and magnetic flux surfaces in the tokamak shown in the diagram below. Discuss how the shape of the q-profile influences plasma stability and confinement.

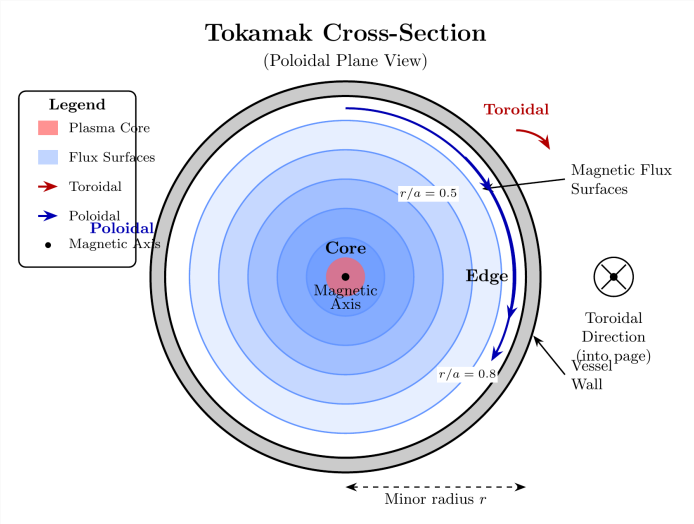

Fig. 4. A diagram of a tokamak showing the magnetic flux surfaces and the q-profile (safety factor). The tokamak should be depicted as a toroidal structure with concentric magnetic flux surfaces. The q-profile should be represented as a curve or graph overlaying the magnetic surfaces, indicating how the safety factor changes across the radius of the tokamak. Include labels for the core, edge, and magnetic axis of the tokamak. IMPORTANT: The diagram is missing required labels. The diagram is missing the q-profile, which. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram is missing the q-profile, which is. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 5 10.0 points

Examine the diagram below illustrating plasma instability growth rate as a function of wave number. Discuss the implications of this relationship for tokamak operation and stability control.

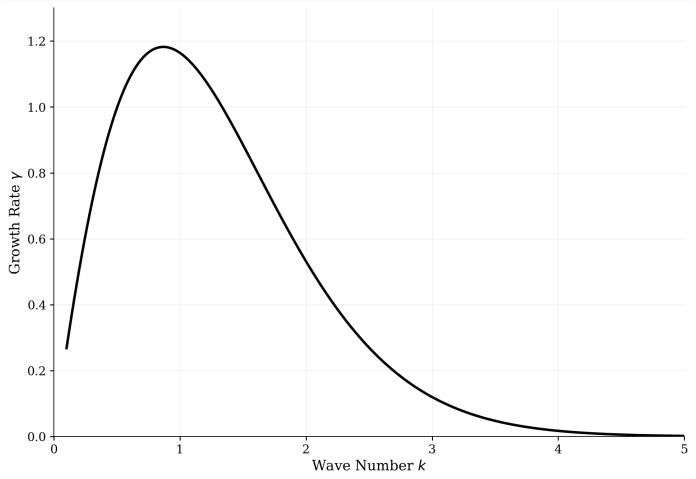

Fig. 5. A plot showing plasma instability growth rate as a function of wave number. The x-axis represents the wave number, and the y-axis represents the growth rate. The plot should illustrate how the growth rate varies with changes in wave number, which is relevant for understanding tokamak operation and stability control.